

Post-Interview Assignment

Cyber Incident Investigation Intelligence — Round 2

TOOLS	SUBMISSION
Use any tools you have access to, including AI coding assistants	Bring to Interview 2

Background

A healthcare organisation has engaged your team following a suspected intrusion. Their SIEM has been collecting events across endpoints, authentication systems, network firewalls, and cloud storage for the past 72 hours. The security team believes an attacker gained an initial foothold via a spear-phishing email, but the full scope — what was accessed, how the attacker moved laterally, and whether data was exfiltrated — is still unknown.

The CISO needs answers fast: which systems were compromised, what did the attacker do, and what is the blast radius? The current process requires a senior analyst to spend 6-8 hours manually correlating thousands of log lines across five different source systems before they can even begin to piece together a timeline.

A previous attempt used a basic log search tool. It returned raw results with no correlation across sources, no timeline reconstruction, and no connection to known attack techniques. Analysts described it as "grep with a UI." Leadership lost confidence in it after it missed a lateral movement event that was plainly visible in hindsight.

The bar is not "can it search logs." The bar is "can the CISO hand this reconstructed attack timeline to their board and legal team with confidence?"

You are coming in as an Agent/GenAI Developer and Applied AI Engineer. The client wants to understand whether modern agentic workflows can materially accelerate incident investigation. They have asked you to build a lightweight prototype that demonstrates the direction.

Your Task

Using the provided synthetic SIEM log dataset (siem_logs.json, included with this assignment) and the publicly available MITRE ATT&CK framework, design and build an artifact that helps an analyst investigate the incident, reconstruct the attack timeline, and answer questions about scope and impact.

At minimum, your system should attempt to:

- **Multi-source log correlation** — Ingest and correlate events across multiple log source types: endpoint, authentication, network, and cloud storage.
- **Attack timeline reconstruction** — Produce a chronological attack timeline from the raw events, identifying the initial access vector, privilege escalation, lateral movement, and potential exfiltration steps.
- **MITRE ATT&CK; TTP mapping** — Map observed behaviours to MITRE ATT&CK tactics and techniques (<https://attack.mitre.org>). Cite the specific technique ID and name for each mapped event.
- **Blast radius assessment** — Identify which hosts and accounts were involved, estimate the blast radius, and flag any gaps or uncertainty in the reconstruction.
- **Show full traceability** — Cite the specific log event(s) — source, timestamp, event ID — behind every claim. A correct "I can see suspicious activity but cannot confirm exfiltration because the relevant logs are absent" beats a

fabricated conclusion.

- **Demonstrate an agentic workflow** — Ingest → parse → correlate → enrich with ATT&CK → synthesise → answer. Tool use, structured intermediate representations, multi-step reasoning, and verification passes are all fair game.

Format is flexible: chat UI, CLI, notebook, agent demo, or lightweight web app. Use whatever format best communicates your approach.

Data Sources

Provided synthetic data (included with this assignment):

- **siem_logs.json** — approximately 300 synthetic log events across four source types (endpoint, auth, network, cloud_storage), spanning a 72-hour window. The dataset contains a realistic multi-stage attack embedded within normal background noise. Source types, event schemas, and field names are described in the file's metadata section.

Public data (no authentication required):

- **MITRE ATT&CK STIX feed** — full enterprise technique catalogue as machine-readable JSON: <https://raw.githubusercontent.com/mitre/cti/master/enterprise-attack/enterprise-attack.json> Alternatively, use the ATT&CK website directly at <https://attack.mitre.org>.

Reasonable scope reductions are encouraged. You may focus on a subset of log source types, map only the most clearly evidenced ATT&CK techniques, or limit your timeline to the confirmed attack path. Strong candidates make thoughtful cuts and are explicit about what they excluded and why.

Example Queries to Test Against

At minimum, your artifact should attempt to handle these:

- "Walk me through the full attack timeline from initial access to last observed activity."
- "What was the initial access vector and what evidence supports that conclusion?"
- "Which MITRE ATT&CK techniques did the attacker use? List them with technique IDs."
- "Which user accounts were compromised or used by the attacker?"
- "Which internal hosts did the attacker move to after the initial foothold?"
- "Is there evidence of data exfiltration? If so, what was accessed and when?"
- "What is the blast radius — list every affected host and account."
- "Where are the gaps in our log coverage that limit your confidence in this reconstruction?"
- "Give me a 3-sentence executive summary suitable for a board briefing."
- "What should the incident response team do in the next 2 hours to contain this?"

Deliverables

For your next interview, bring:

1. **A tangible artifact** — prototype application, notebook, architecture diagram, demo workflow, or CLI. Rough is fine. Something tangible is not optional.
2. **A 5–10 minute walkthrough** — what you built, architectural decisions, where AI/agents were used, where they were intentionally not used, what shortcuts you took, and what you would improve with another day or week.

3. **A short written or verbal discussion** covering: trust & hallucination (how does your approach reduce fabrication risk in an investigation context, and where can it still fail?), agentic workflow design (why did you choose your structure?), and product thinking (what type of security user is this best suited for today, and what would it take to make it production-ready?).

What We Are Evaluating

Not grading on polish. Here is what we are actually looking at:

- **Applied reasoning** — Can you design a practical investigation system around an ambiguous, noisy, real-world dataset?
- **Engineering judgment** — Did you make sensible tradeoffs under time constraints? Can you articulate them clearly?
- **Agentic thinking** — Can you structure a multi-step agentic workflow that combines log parsing, cross-source correlation, external knowledge (ATT&CK), and natural-language synthesis?
- **Trust & traceability** — Do you understand why fabricated conclusions are especially dangerous in an incident investigation, and how your approach addresses this?
- **Communication** — Can you clearly explain what your system found, what it is uncertain about, and why?
- **Product intuition** — Did you think about the real end user — a SOC analyst or CISO — and what they actually need under pressure?

AI-assisted implementation is expected and encouraged. How you use AI tooling is part of what we are evaluating.